Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 16: Relations That Behave Like Equality

Equivalence Relations

Chartrand & Zhang, Section 5.2

Name: _____ Date: _____

Where we are

The best-known relation on $\mathbb{Z}$ is $=$. It is reflexive, symmetric, and transitive. Many relations in mathematics and computing *mimic* equality in exactly those three respects without being equality — and those are worth a name.

By the end of today you should be able to

1. State the definition of an equivalence relation.
2. Prove that a given relation is one, in three parts.
3. Recognise equivalence relations you have already been using.

1. The definition

Definition 5.14

A relation R on a nonempty set is an **equivalence relation** if R is **reflexive**, **symmetric**, and **transitive**.

Three properties means three proofs

A proof that R is an equivalence relation has **three separate parts**, each with its own opening and closing. Two out of three proves nothing — exactly as one half of a biconditional proved nothing in Lecture 6.
Label the parts. A grader should be able to see where each begins.

2. Ones you have been using all term

Relation	Why it is an equivalence relation
$=$ on $\mathbb{Z}$	The three properties are basic facts about equality.
$\equiv$ on compound statements	$P \equiv P$; if $P \equiv Q$ then $Q \equiv P$; equivalences chain.
Parallel-or-equal on lines	Example 5.15 — every line is parallel to itself, and so on.

Week 1 was an equivalence relation all along

Logical equivalence $\equiv$ is reflexive, symmetric, and transitive. That is why you could chain rewrites — De Morgan, then double negation, then commutativity — and still be talking about the same statement at the end.
You relied on transitivity every time you wrote a derivation. Today it gets its name.

3. Worked proof: $a\ R\ b$ if $a+b$ is even

Claim (Example 5.21)

Define R on $\mathbb{Z}$ by $a\ R\ b$ if $a+b$ is even. Then R is an equivalence relation.

Reflexive. Let $a \in \mathbb{Z}$. Then $a + a = \underline{2a}$, which is even. So $a\ R\ a$.

Symmetric. Assume $a\ R\ b$, so $a+b$ is even. Since $b + a = \underline{a+b}$, it follows that $b+a$ is even, so $b\ R\ a$.

Transitive. Assume $a\ R\ b$ and $b\ R\ c$. Then $a+b$ and $b+c$ are both even, so $a+b=2x$ and $b+c=2y$ for some integers x, y. Adding,

$$(a+b)+(b+c) = 2x+2y \qquad \text{so} \qquad a+c = 2x+2y-2b = 2\big(\underline{x+y-b}\big).$$

Since that quantity is an integer, $a+c$ is even, so $a\ R\ c$. ■

The move worth stealing

In the transitive part we wanted $a+c$ but were handed $a+b$ and $b+c$. **Adding the two hypotheses made b cancel** and produced exactly what was needed.
When a proof gives you two facts and asks for a third, try combining them and see what survives.

4. Your turn (start it here, finish on paper)

Claim

Define R on $\mathbb{Z}$ by $a \; R \; b$ if $a - b$ is even. Show R is an equivalence relation.

Reflexive. $a - a = 0$, which is even.

Symmetric. If $a - b = 2k$ then $b - a = -2k$, also even.

Transitive. If $a - b = 2k$ and $b - c = 2m$, then $a - c = (a - b) + (b - c) = 2(k + m)$.

5. When it fails

Relation on $\mathbb{Z}$	Equivalence?	What fails
$a \; R \; b$ if $a \leq b$	No	Not symmetric: $1 \leq 2$, $2 \not\leq 1$.
$a \; R \; b$ if $ab \geq 0$	No	Not transitive: $-1, 0, 1$.
$a \; R \; b$ if $ab > 0$	No	Not reflexive: $0 \cdot 0 \not> 0$.
$a \; R \; b$ if $\lvert a \rvert = \lvert b \rvert$	Yes	—

One failure is enough

To show R is *not* an equivalence relation you need only one property to fail, with a witness. You do not have to check the other two.
That is the universal-versus-existential asymmetry from Week 4, for the fourth time.

Exit ticket

1. Name the three properties an equivalence relation must have.

2. Is $\subseteq$ on $\mathcal{P}(\{1, 2\})$ an equivalence relation? Justify with a witness if not.

Before Friday

Read Section 5.2, pp. 153–158 — equivalence classes (Definition 5.17) and Theorem 5.23.

Self-check (not collected). Section 5.2, Exercises 1, 3, and 7 — odd-numbered, answers in the back.

Homework 6 is due Friday, September 25, 11:59 p.m.

Questions I still have